Course Description

Biology Courses

BI201	Introduction to Modern Biology	
	Credit Hours: 3	Contact Hours (L, H): (3, 0)
	Prerequisites: None	

Characteristics of life, cell structure and function, molecular basis of life. Cell biology: enzymes, proteins, nucleic acids. Membrane transport. Bioenergetics and metabolism. Information storage and processing. Reproduction and patterns of inheritance. Genetics. Theory of Evolution, microevolution and macroevolution.

Business Management Courses

BM111	Introduction to Management	
	Credit Hours: 3	Contact Hours (L, H): (3, 0)
	Prerequisites: None	

Management components: Planning, Organizing, Leading, Controlling. Structure of organizations. Introduction to all functional areas of management: Accounting, Finance, Marketing, Production, Human Resources and Information Technology. Business strategy and policies. Introduction to strategy formulation.

BM201	Financial and Managerial Accounting	
	Credit Hours: 3	Contact Hours (L, H): (3, 0)
	Prerequisites: MA101	

Basic accounting concepts, practices and procedures. Standards and practices in preparation and communication of financial data and information relevant to decision-making. Accounting cycle. Financial statements. Assets, equity, liability (both long term and short term). Cash flows. Analysis of financial statements. Measuring and estimating costs, budgeting and cost analysis in business planning and decision-making.

BM223	Managing Human Resources	
	Credit Hours: 3	Contact Hours (L, H): (3, 0)
	Prerequisites: BM111	

Analysis of human resource requirements in organizations. Recruitment and selection. Training and development, team building, leadership and appraisal. Interpersonal relationships and dynamics. Division of responsibilities, delegation, outsourcing and consultants. Wages and salary administration, benefits. Legal requirements. Safety and security. Cost-benefit analysis. Employee assistance. Quality of work-life.

BM227	Leadership Competencies for Engineers	
	Credit Hours: 3	Contact Hours (L, H): (3, 0)
	Prerequisites: BM111	

The objective of this course is to focus on the basic principles of personal and interpersonal leadership that should be part of the skills-set of today's scientists and engineers. It is a case-study based course, with an emphasis on vision, objectives, motivation, decision-making, time management, team building, conflict, and ethics, dealing with change, communication skills, organizational politics and diversity issues. Teaching methods include lectures, class discussions, videos, oral presentations, written assignments, and group projects. By the end of the course, students are expected to have gained better personal and interpersonal awareness, as well as a greater understanding of the complex organizational issues facing scientists and engineers in their daily work.

BM228	Strategic Management	
	Credit Hours: 3	Contact Hours (L, H): (3, 0)
	Prerequisites: BM111	

Strategy concept and the strategic management process. Purpose of the organization: vision and mission statements, corporate objectives and stakeholder analysis. Evaluating a firm's external environment. Evaluating a firm's external environment. Evaluating a firm's internal capabilities Strategic positioning: cost leadership and product differentiation. Vertical integration. Corporate Diversification. Strategic alliance and mergers & acquisitions.

BM229	Project Management	
	Credit Hours: 3	Contact Hours (L, H): (3, 0)
	Prerequisites: BM111	

As organizations continue to look for ways to reduce costs, engineers are often expected to manage the execution of projects, in addition to their other responsibilities. Project management is about managing and leading complex projects and includes the following topics of study: scope management, time management, cost management, human resources management, communications, quality management, and professional responsibility (i.e., ethics). In this course, scientists and engineers will learn how to break down a complex project into manageable segments, lead a diverse project team, and use effective tools to ensure that the project meets its promised deliverables and is completed within budget and on schedule. The course is practical in nature, making intensive use of Microsoft Project software.

BM230	Introduction to Industrial Management	
	Credit Hours: 3	Contact Hours (L, H): (3, 0)
	Prerequisites: BM111	

Industrial Management (IM) is the organizational process that includes strategic planning, objective setting, managing resources, deploying the human and financial assets needed to achieve objectives, and measuring results. The aim of the course is to introduce students to the key issue of reaching maximum output with minimum costs of production while observing principles of Quality Management (QM).

BM301	Innovation and Entrepreneurship	
	Credit Hours: 3	Contact Hours (L, H): (3, 0)
	Prerequisites: BM111	

Developing a comprehensive business plan. Operating management of small companies. Venture capital and financing. Issues and options in developing a new technological venture. Identification of opportunities, evaluation of technical feasibility and commercial potential, planning for commercialization. Forecasting and assessing technologies.

BM421	Business Marketing and Sales	
	Credit Hours: 3	Contact Hours (L, H): (3, 0)
	Prerequisites: BM111, BM227	

This course builds on the general knowledge acquired about the business process from previous courses. It further investigates the subject of Marketing & Sales and provides a comprehensive overview of the elements involved. The course focuses on both strategic theory and field implementation aspects of managing marketing issues in business. Relevant case discussions and field projects which link strategy with sales and objectives are used throughout the synchronous sessions, with particular interest in both B2B & B2C marketing.

Computer Engineering Courses

CE100	Introduction to Computer Systems	
	Credit Hours: 3	**Contact Hours (L, H): (2, 2)**
	Prerequisites: None	

This course covers basic topics in computer systems hardware and software. This includes an overview of the main components of a computer (hardware) and elementary concepts of programming (using Python), students learn how to write simple programs that including variables, types, Control statements – conditions & iterations, arrays (1D and 2D) and basic operation and introduction to functions. It is intended for students without a background in any programming language. Finally give students an overview of Raspberry Pi computer – setup and prepare the computer, basic exercises with breadboard LED circuits. Introduction to the Arduino microcontroller – simple control tasks including LED blinking, switch-based control of LEDs.

CE212	Digital Logic Design	
	Credit Hours: 4	**Contact Hours (L, H): (3, 2)**
	Prerequisites: CS101	

This course provides a modern introduction to logic design and the basic building blocks used in digital systems digital computers. It starts with a discussion of combinational logic: logic gates, minimization techniques, arithmetic circuits, and modern logic devices such as field programmable logic gates. The second part of the course deals with sequential circuits: flip-flops, synthesis of sequential circuits, and case studies, including counters, registers, and random-access memories. State machines will then be discussed and illustrated through case studies of more complex systems using programmable logic devices. Different representations including truth table, logic gate, timing diagram, switch representation, and state diagram will be discussed.

CE222	Computer Architecture	
	Credit Hours: 4	**Contact Hours (L, H): (3, 2)**
	Prerequisites: CE212	

This course will describe the basics of modern processor operation. Topics include computer system performance, instruction set architectures, pipelining, branch prediction, memory-hierarchy design, and a brief introduction to multiprocessor architecture issues. The course includes practical lab sessions to reinforce theoretical concepts.

CE334	Computer Network Protocols and Applications	
	Credit Hours: 3	**Contact Hours (L, H): (3, 0)**
	Prerequisites: CE222	

This course will describe the basics of modern processor operation. Topics include computer system performance, instruction set architectures, pipelining, branch prediction, memory-hierarchy design, and a brief introduction to multiprocessor architecture issues. The course includes practical lab sessions to reinforce theoretical concepts.

CE364	Human-Machine Interaction	
	Credit Hours: 3	Contact Hours (L, H): (2, 2)
	Prerequisites: CS203, CS361	

Introduction to fundamental methods and principles for designing, implementing, and evaluating user interfaces. Topics: user-centered design, rapid prototyping, experimentation, direct manipulation, cognitive principles, visual design, social software, software tools. Depth work includes ubiquitous computing, social computing and design plus creation. Breadth work includes HCI methods, programming, visualization and user modelling. Lab will place emphasis on interaction design process, needs analysis, user observation, sketching, concept generation, scenario building and evaluation.

CE376	Mobile Applications Development	
	Credit Hours: 3	Contact Hours (L, H): (2, 2)
	Prerequisites: CS203	

This course covers the fundamentals of mobile application life cycle and platforms, Design and develop of cross platform mobile application using Xamarin. Use XAML for User Interface (UI) design, apply code-behind programming using C# language. Understand MVVM (Model-View-ViewModel). Connect to Database and publish the mobile application.

CE401	Self-Study	
	Credit Hours: 2	Contact Hours (L, H): (0, 4)
	Prerequisites: None	

In-depth study of material not ordinarily covered in a course, under the supervision of a faculty member. Emphasis is on comprehension and presentation and survey of a clearly identified sub-area of computer engineering.

CE402	Minor Project	
	Credit Hours: 3	Contact Hours (L, H): (0, 6)
	Prerequisites: None	

Provides students with an opportunity for developing and implementing a system.

CE403	Minor Project	
	Credit Hours: 2	Contact Hours (L, H): (0, 4)
	Prerequisites: None	

Provides students with an opportunity for developing and implementing a system.

CE404	Minor Project	
	Credit Hours: 1	Contact Hours (L, H): (0, 2)
	Prerequisites: None	

Provides students with an opportunity for developing and implementing a system.

CE420	Advanced Computer Architecture	
	Credit Hours: 2	Contact Hours (L, H): (3, 0)
	Prerequisites: CE222	

Classification of parallel computing structures. Principles of pipelined computer systems. Super scalar and VLIW architectures. SIMD multiprocessor structures. Interconnection networks. MIMD issues in multiprocessor systems. Compiler issues for parallel architectures. High performance memory system. Embedded Systems.

CE421	Embedded Systems	
	Credit Hours: 3	Contact Hours (L, H): (2, 2)
	Prerequisites: CE222	

This course introduces designing, interfacing, configuring, and programming embedded systems. The course contents include embedded system architecture, hardware-software interfaces, memory architecture, software design, and enabling components in embedded systems such as clocks, general purpose I/Os, interrupts, busses, analog to digital and digital to analog converters. The lectures in the course will be complemented by laboratory sessions where students will develop embedded computing system applications

CE491	Major Project, Part 1	
	Credit Hours: 4	Contact Hours (L, H): (0, 8)
	Concurrent Prerequisites: GE480 and CS391 Minimum number of completed credit hours:90	

This project-based course represents the first half of a project that requires students to demonstrate technical and practical skills and experience in some aspects of computer science/engineering, communications, and electronics engineering, as well as personal attributes at levels which commensurate with professional and ethical engineering practices. Students (normally a group of 3 supervised by a KCST Supervisor) are normally assigned a project. Then they are expected to state its objectives, complete a literature survey, perform a feasibility study, set project specifications, and select a design method. Preliminary modeling and analysis are also expected from students to ensure that the necessary material needed for the completion of the project can be acquired in time for the completion of the project in the spring term. Teams will submit a professional report and do the relevant oral presentation. This course sequence prepares students for entry to the workplace. Deliverables include a Project Plan Document, a Design Document, and an oral presentation of project achievements.

CE492	Major Project, Part 2	
	Credit Hours: 6	Contact Hours (L, H): (0, 12)
	Prerequisites: CE491	

This project-based course is a continuation of CE491 (FYPI). Students are asked to deliver a product that has passed through the design, analysis, testing and evaluation stages. Verification and validation must be applied at all stages, and implementation and testing must reflect that the design specifications including identified engineering ethics have been satisfied. Students are also expected to submit a final professional report that includes a description of the design process, implementation and testing, verification and validation and a critical appraisal of the project. A Final Report, an Oral Presentation of project achievements, and a Poster are also part of the project deliverables.

Chemistry Courses

CH101	Physical Chemistry	
	Credit Hours: 3	Contact Hours (L, H): (2, 2)
	Prerequisites: None	

Chemical thermodynamics: Free energy and entropy changes, phase rule and phase equilibria. Chemical dynamics: Stoichiometry of reactions, rate of chemical reactions, surface phenomena (absorption, catalysis), homogeneous and heterogeneous catalysis. Electrochemistry: Equilibrium electrochemistry. Quantum mechanical principles of structure and bonding in molecules. Bond energy, energy state of molecules.

Computer Science Courses

CS100	Introduction to Computing	
	Credit Hours: 4	Contact Hours (L, H): (3, 2)
	Prerequisites: None	

The course aims at introducing the students to the computer science and main engineering concepts. This includes the main components of a computer, numbering systems, problem-solving and the logic of converting real-world problems into algorithms. To that end, the course covers basic language syntax, simple data types, variables, decision structures, repetition structures, arrays, strings and debugging tools. It is intended for students with a prior background in any programming language.

CS101	Introduction to Computers & Programming	
	Credit Hours: 4	Contact Hours (L, H): (3, 2)
	Prerequisites: CE100 or CS100	

This course covers functions calls and returns, parameter passing by value and by reference, and basic File operations. The course also covers searching and sorting techniques in 1D Arrays, using -2D arrays, and pointers manipulation.

CS112	Data Abstraction & Object-Oriented Programming	
	Credit Hours: 4	Contact Hours (L, H): (3, 2)
	Prerequisites: CS101	

Review of elementary structured data types (arrays and structures) and operations on them. Introduce classes, objects, methods, private and public members, constructors and destructors and overloading. Illustrate object-oriented concepts of data abstraction, inheritance, polymorphism and virtual functions. It is intended for students with a prior background in any programming language.

CS193	Software Tools	
	Credit Hours: 3	Contact Hours (L, H): (2, 2)
	Prerequisites: None	

The course deals with tools for creating and managing software and data. Editors and structured programming environments. Program documentation and literate programming tools. Version control software. Spreadsheets, word processors, presentation tools. Configuration management, browsers and search tools. Software installation. Modeling and simulation techniques. Graphical user interfaces

CS201	Critical Thinking and Problem Solving	
	Credit Hours: 2	Contact Hours (L, H): (2, 0)
	Prerequisites: None	

The course will develop students' distinction between illogical and logical reasoning and strong and weak arguments. Students will learn to define problems and questions clearly and precisely and to identify assumptions behind statements, claims and arguments. Students will also learn to use strategies for effective problem solving and apply these to a range of practical problems by learning how to deconstruct statements and understand the underlying ideology or prejudice underpinning some beliefs. The course includes problem-solving techniques used in Computer Science.

CS203	Data Structures and Objects	
	Credit Hours: 4	Contact Hours (L, H): (3, 2)
	Prerequisites: CS101	

This course introduces classes, objects, methods, private and public members, constructors and destructors and overloading. The course also introduces students to the study of data, its representation and processing by computer systems. The course enables learners to use data structures to design algorithmically efficient computer programs. Topics covered in this course includes list ADT, array-based lists, linked lists, stacks, queue ADT, binary trees, binary search trees, AVL trees, graphs, and their applications. The course also introduces the concept of algorithm analysis to evaluate the efficiency attained in algorithms used to manipulate data structures. Function templates, class templates, and Standard Template Library (STL) in C++ are also introduced. This course is accompanied with hands-on lab sessions to help students further understand the various topics.

CS221	Computer Organization	
	Credit Hours: 4	Contact Hours (L, H): (3, 2)
	Prerequisites: CS212	

Information representation and binary arithmetic. Basic combinational and sequential circuit design. Structures and subsystems of a computer. Instructions and their formats. Assembly programming. CPU organization and ALU design. Bus organization and control. Memory organization and hierarchy. I/O structures, interrupt, DMA. Examples of organization of different classes of computers.

CS301	Numerical and Scientific Computing	
	Credit Hours: 3	Contact Hours (L, H): (2, 2)
	Prerequisites: CS203	

Numerical solution of linear and non-linear systems. Differentiation, integration, curve fitting, interpolation. Numerical optimization: maxima, minima of non-linear functions. Linear and non-linear programming. Eigenvalue problems, singular value decomposition. Numerical methods for differential equations, initial value problems and boundary value problems.

CS304	Numerical Computation Methods	
	Credit Hours: 3	Contact Hours (L, H): (2, 2)
	Prerequisites: CS203, MA203	

Error analysis: forward, backward, rounding, floating point errors. Review of matrices and linear systems. Eigenvalues and singular value decompositions and linear systems sensitivity. Review of convergence of iterative methods. Newton's method. Basic introduction to PDEs and discretization storage schemes. Sparse matrices arising from PDEs, Implementation of direct and iterative methods for sparse matrices. Introduction and use of packages: SPARSE PACK, LINPACK, EISPACK and LAPACK, ITPACK and Lacazos methods.

CS322	Operating Systems	
	Credit Hours: 4	Contact Hours (L, H): (3, 2)
	Prerequisites: CE222	

This course teaches basic operating system abstractions, mechanisms and their implementations. Core concepts covered will include processes and threads, CPU job scheduling, synchronization, memory management, file system organization, interrupts handling, input and output device management, security. The course also covers lower-level interaction details between operating system, Input/output (I/O) devices, memory components, and user applications in different conditions. The lab sessions will help students understand the implementation details related to process management techniques, job scheduling algorithms, memory management and file allocations techniques. Students will also acquire basic knowledge of different system and I/O calls in the UNIX operating system environment.

CS333	Computer Network Systems and Protocols	
	Credit Hours: 3	Contact Hours (L, H): (3, 0)
	Prerequisites: CE222	

This course provides students with a solid understanding of the state of the art in computer network systems and protocols. The topics to be covered include current topics of research and development such as network models, categories of networks, network criteria, physical structures, connecting devices, VLAN, quality of service, an overall measurement of network performance, transmission media, switching, IP addressing and subnetting, network services and error detection and correction. The course also helps the students develop expertise in some areas of computer networks and prepares them to start research work in this area.

CS341	Programming Languages	
	Credit Hours: 3	Contact Hours (L, H): (2, 2)
	Prerequisites: MA111, CS203	

Structure of programming language: syntax, grammars, type systems. Abstract machines and runtime systems. Compilers and translators. Lexical and syntactic analysis, top-down and bottom-up parsing techniques, internal form of source programs. Semantic analysis, symbol tables, error detection and recovery, code generation and optimization. Type checking and static analysis.

CS351	Computer Graphics	
	Credit Hours: 3	Contact Hours (L, H): (2, 2)
	Prerequisites: MA111, CS203	

This course covers topics including graphics systems and models; geometric representations and transformations; introduction to OpenGL; application of graphics programming techniques to design and create computer graphics scenes using 2D/3D transformation; viewing and projections; object and color modelling; illumination modelling; texture mapping; and ray tracing techniques. This course is intended for students with prior background in data structures, mathematics involving matrices, vectors, and C++ programming language.

CS352	Multimedia Systems	
	Credit Hours: 3	Contact Hours (L, H): (2, 2)
	Prerequisites: CS322, EE201	

Multimedia data technologies and I/O devices: sound and audio, image and graphics, animation and video, tools for multimedia development. Compression and decompression, analog and digital representation, encoding and decoding algorithms, lossless and lossy compression. Content protection, watermarking, integrity, authentication, IPR issues.

CS361	Artificial Intelligence	
	Credit Hours: 3	Contact Hours (L, H): (2, 2)
	Prerequisites: CS203, CE222	

This course provides an overview of the main concepts and algorithms underlying the understanding artificial intelligence and will cover several concepts, algorithms, and techniques regarding the artificial intelligence such as Problem-solving, search techniques, knowledge representation and reasoning, Expert Systems and Planning, Rule-based systems and Fuzzy Logic, Supervised Learning, Unsupervised Learning, Reinforcement Learning, Regression, Logistic Regression, Linear Regression, Classification, Neural Network, and Deep Neural Network. This course will use the Python as a programming language to implement all lectures topics.

CS371	Databases	
	Credit Hours: 4	Contact Hours (L, H): (3, 2)
	Prerequisites: CS203, MA111	

This course introduces the fundamental concepts necessary for designing, using, and implementing database systems and database applications. The course stresses the fundamentals of database modeling and design, the languages and models provided by the database management systems, and database system implementation techniques. The course objective is to provide a working knowledge, practices, and up-to-date presentation of the most important aspects of database systems and applications, and related technologies. The course assumes that students are familiar with elementary programming and data structure concepts and that they have had some exposure to the basics of computer organization. The course is accompanied with laboratory sessions where students can build and interact with databases and gain hands-on experience while practicing theoretical concepts they have learned in the classroom.

CS372	Web Information Systems	
	Credit Hours: 3	Contact Hours (L, H): (2, 2)
	Prerequisites: CS203, MA111	

Structure of the World Wide Web. Semi-structured data. HTML, HTTP, XML and translators from XML. Browsers and Search engines. Domain Name Services. Web services.

CS373	Analysis and Design of Algorithms	
	Credit Hours: 3	Contact Hours (L, H): (2, 2)
	Prerequisites: CS203, MA111	

This course presents the fundamental techniques for designing efficient computer algorithms, proving their correctness, and analyzing their running times. General topics include mathematical analysis of algorithms, advanced data structures such as balanced search trees, algorithm design techniques, and graph algorithms. The course requires prior background in discrete mathematics and data structures. The course will have theoretical and laboratory sessions where students will acquire the understanding of basic algorithm design techniques for different problems, and corresponding methodologies for asymptotic analysis. The laboratory session in this course will help students experimentally analyze algorithm behavior with synthetic input.

CS391	Software Engineering	
	Credit Hours: 3	Contact Hours (L, H): (2, 2)
	Prerequisites: CS371	

This course covers topics which include the fundamentals of software engineering, including understanding system requirements, finding appropriate engineering compromises, effective methods of design, coding, and testing, team software development, and the application of engineering tools. The course will combine a strong technical focus with a project providing the opportunity to practice engineering knowledge, skills, and apply best practices in a realistic development setting with a real client.

CS401	Self Study	
	Credit Hours: 2	Contact Hours (L, H): (0, 4)
	Prerequisites: None	

In-depth Study, of material not ordinarily covered in a course, under the supervision of a faculty member. Emphasis is on comprehension, presentation and survey of a clearly identified sub-area of computer science.

CS402	Minor Project	
	Credit Hours: 3	Contact Hours (L, H): (0, 6)
	Prerequisites: None	

Provides students with an opportunity for developing and implementing a system.

CS403	Minor Project	
	Credit Hours: 2	Contact Hours (L, H): (0, 4)
	Prerequisites: None	

Provides students with an opportunity for developing and implementing a system.

CS404	Minor Project	
	Credit Hours: 1	Contact Hours (L, H): (0, 2)
	Prerequisites: None	

Provides students with an opportunity for developing and implementing a system.

CS413	Theory of Computation	
	Credit Hours: 3	Contact Hours (L, H): (3, 0)
	Prerequisites: CS203, MA111	

Models of computation. Finite state automata and regular languages. Context-free grammars and push-down automata, Turing machines, other equivalent models of computation, computability and undecidability. Introduction to recursive function theory. Church's thesis. Complexity classes.

CS426	Distributed Computation	
	Credit Hours: 3	Contact Hours (L, H): (3, 0)
	Prerequisites: CS322	

Models of distributed computation. Synchronous and asynchronous models. Fundamental problems and algorithms: consensus, leader election, snapshot, termination detection, fault tolerance and stabilization. Transactions, serializability, atomicity.

CS439	Information Security	
	Credit Hours: 3	Contact Hours (L, H): (2, 2)
	Prerequisites: CS203	

Network security, firewalls, sandboxes, Hardware protection. Cryptography. Secret key and public key algorithms, digital signatures, privacy, authentication. Dolev-Yao intruder model. Covert channels, information flow, Bell-LaPadula model. Threats, viruses, worms. Access control. Computer crime. Professional and ethical issues. Legal issues.

CS491	Major Project, Part I	
	Credit Hours: 4	Contact Hours (L, H): (0, 8)
	Concurrent Prerequisites: CS391, GE480 Minimum number of completed credit hours: 90	

This project-based course represents the first half of a project that requires students to demonstrate technical and practical skills and experience in some aspects of computer science/engineering, communications, and electronics engineering, as well as personal attributes at levels which commensurate with professional and ethical engineering practices. Students (normally a group of 3 supervised by a KCST Supervisor) are normally assigned a project. Then they are expected to state its objectives, complete a literature survey, perform a feasibility study, set project specifications, and select a design method. Preliminary modeling and analysis are also expected from students to ensure that the necessary material needed for the completion of the project can be acquired in time for the completion of the project in the spring term. Teams will submit a professional report and do the relevant oral presentation. This course sequence prepares students for entry to the workplace. Deliverables include a Project Plan Document, a Design Document, and an oral presentation of project achievements.

CS492	Major Project, Part II	
	Credit Hours: 6	Contact Hours (L, H): (0, 12)
	Prerequisites: CS491	

This project-based course is a continuation of CS491 (FYPI). Students are asked to deliver a computing-based product or service that has passed through the problem analysis, solution design, testing and evaluation stages. Verification and validation must be applied at all stages, and implementation and testing must reflect that the solution requirements including identified professional and scientific ethics have been fulfilled. Students are also expected to submit a final technical report that discusses the detailed description of the design process, implementation and testing, verification and validation and a critical appraisal of the project. A final report, an oral presentation of project achievements, and a poster are expected deliverables towards the end of the project lifetime.

Electronics and Communications Engineering Courses

EE101	Introduction to Electrical Engineering	
	Credit Hours: 4	Contact Hours (L, H): (3, 2)
	Prerequisites: None	

This course covers topics which include Introduction to electric circuits, fundamental laws in electrical circuits such as Ohm's law and Kirchhoff's laws. The DC circuit analysis techniques such as nodal analysis, mesh analysis techniques, and network theorems. First order transient analysis of RC and RL circuits. AC steady-state analysis: Basic characteristics of AC waveforms- frequency, amplitude and phase, Peak and Root mean square values.

EE112	Electronic Instrumentation	
	Credit Hours: 3	Contact Hours (L, H): (2, 2)
	Prerequisites: EE101	

This course aims to develop an understanding of the function of the first key parts of a typical instrumentation system: Sensors, signal conditioning, sampling, instrumentation amplifiers, filters, sources of noise and reduction. Diodes and Applications, Bipolar junction, Field effect and MOS transistors, Small-signal equivalent model of transistors and voltage amplifiers, Operational Amplifiers, Regulated power supplies, Introduction to instruments, Static and dynamic performance of instruments, Electrical measurements. Projects dealing with the design of electronic circuits, implementation on breadboards and testing.

EE201	Signals and Systems	
	Credit Hours: 4	Contact Hours (L, H): (3, 2)
	Prerequisites: MA102	

Motivation and applications. Signal classifications and operations. System characterizations, linearity, time invariance, causality, stability, memoryless, invertibility. Convolution and LTI system responses. Fourier series and transform for continuous time signals. Laplace transform with applications to commercial systems. Introduction to discrete time signals and systems.

EE211	Circuit Theory	
	Credit Hours: 4	Contact Hours (L, H): (3, 2)
	Prerequisites: MA101, EE101	

This course covers topics which include introduction to Sinusoidal analysis, Phasor relationships for circuit elements, Series/parallel impedances, Nodal and Mesh Analysis for AC circuits; Instantaneous and average AC power; RMS value, Complex AC power, and power factor computation; Three-phase circuits analysis, power in balanced three-phase system; Filters, frequency response; resonant electric circuits, and different types of responses of two energy storage elements in electric circuits.

EE 212	Digital Electronics	
	Credit Hours: 4	Contact Hours (L, H): (3, 2)
	Prerequisites: EE101	

This course covers topics which include Digital representation of information, Digital design using Complementary Metal Oxide Semiconductor (CMOS) logic circuits, CMOS Inverter, Static Combinational circuits, Dynamic Combinational circuits, Static Sequential circuits, Dynamic Sequential circuits, Semiconductor Memories. Practical and Lab activities: MOSFET DC Characteristics, Measure static and dynamic characteristics of the CMOS Inverter, Basic CMOS Gates (NAND, AND, NOR and OR), Boolean function implementation using standard CMOS circuits, Sequential CMOS Circuits (Flip Flop and Latch).

EE312	Electronic Devices and Systems	
	Credit Hours: 4	Contact Hours (L, H): (3, 2)
	Prerequisites: EE212	

This course aims to introduce the basic concepts and design of analog electronics circuits and it covers the topics which include, review of current flow mechanisms in BJT and MOSFET, principles of operation of BJT and MOSFET amplifiers, Biasing of BJT and MOSFET amplifier, Integrated circuit amplifier, differential amplifiers and multistage amplifiers. Finally, this course discusses the concepts of frequency responses in BJT and MOSFET amplifiers.

EE341	Communication Systems	
	Credit Hours: 4	Contact Hours (L, H): (3, 2)
	Prerequisites: EE201	

This course introduces analogue communication systems, with associated signal analysis methods, design engineering principles, and the actual communication systems implementation/experimentation based on National Instruments-Universal Software Radio Peripherals (USRPs). Learners enrolled in this course, will be able to conceive, design, implement and operate complex engineered communication systems. A learner is required to have strong background of fundamental mathematical models prescribed in the Signals and Systems course undertaken previously. Key topics included in this course include – Introduction to Communication systems, Frequency Domain Analysis, Modulation and Communication channels, amplitude/linear Modulation, e.g., double sideband (DSB), single sideband (SSB), amplitude modulation (AM), Quadrature AM (QAM), angle modulation, e.g., frequency modulation (FM) and phase modulation (PM), Sampling and Reconstruction.

EE342	Data Communications	
	Credit Hours: 4	Contact Hours (L, H): (3, 2)
	Prerequisites: EE201, MA301	

This course introduces the data processing in Data Communication Systems. In particular, following issues will be covered; (1) Source Coding: General Concepts, Expected Length, Uniquely Decodable Codes, Prefix Codes, and Huffman Coding. (2) Extension Coding, Entropy, Convergence to Entropy. (3) Digital Communication Systems over Discrete Memoryless SS Channel (DMC): DMC, Optimal Detection for DMC, Binary Symmetric Channel, Binary Asymmetric Channel, Symbol Error Probability. (4) Maximum a Posteriori Probability (MAP) detector, Maximum Likelihood (ML), Block Encoding, Minimum-Distance Decoder, Hamming Distance. (5) Mutual Information. (6) Channel Capacity. (7) Channel Coding: Linear Block Codes, Generator Matrix, Parity Check Matrix. (8) Hamming Codes, Interleaving, Binary Convolutional Codes.

EE351	Control Systems and Instrumentation	
	Credit Hours: 4	Contact Hours (L, H): (3, 2)
	Prerequisites: EE201	

This course covers topics which include introduction to different types of transducers and their applications, Modeling in Time and Frequency domains, State space representation, Block diagrams, Laplace transform, Time response, Steady state error and transient response, Open and Closed-loop control systems, Stability of feedback systems, Routh-Hurwitz Stability, Root Locus Method, Introduction to PID controllers, Frequency response.

EE352	Digital Communications	
	Credit Hours: 4	Contact Hours (L, H): (3, 2)
	Prerequisites: EE341	

This course introduces digital communication systems and will cover the following: (1) Introduction to pulse modulations, e.g., pulse amplitude modulation (PAM), pulse width modulation (PWM), pulse position modulation (PPM), pulse code modulation (PCM). (2) Signal constellations, of PAM, phase shift keying (PSK), frequency shift keying (FSK), PPM, QAM, PWM. (3) Bandwidth of the M-ary and binary transmitted signal. (4) Bandwidth of PAM, QAM and FSK signals. (5) Receivers in digital communication systems, basic concepts, Minimum Euclidean distance receiver, Matched filter receiver, Performance binary signaling. (6) Carrier modulation techniques, Equivalent baseband model, description Intersymbol interference (ISI), eye diagram, pulse shaping (7) Nyquist condition, Spectral raised cosine, Equalizers. The student will have hands-on experience in Lab to design and experiment with the techniques with Universal Software Radio Peripherals (USRPs).

EE360	Introduction to Satellite Engineering	
	Credit Hours: 3	Contact Hours (L, H): (3, 0)
	Prerequisites: EE201	

Space craft and its subsystems, -Orbital classifications, spacing, and frequency allocation, Link models and equations, space segment, Launch vehicle, Payload and Spacecraft Design, Attitude and Orbit Control.

EE362	Electromagnetic Waves and Theory	
	Credit Hours: 3	Contact Hours (L, H): (2, 2)
	Prerequisites: PH102, MA204	

Review of Maxwell's equations, wave propagation in unbounded medium. Boundary conditions, reflection, and refraction of plane waves. Transmission Lines: distributed parameter circuits, travelling and standing waves, impedance matching, Smith chart, analogy with plane waves. Waveguides: parallel-plane guide, TE, TM and TEM waves, rectangular and cylindrical waveguides, resonators. Planar transmission lines: strip-line, micro-strip-line, application of numerical techniques. Dielectric guides and optical fibers. Radiation: retarded potentials, Hertzian dipole, short loop, antenna parameters. Radio-wave propagation: ground-wave, sky-wave, space-wave.

EE363	Analog and Mixed Analog Integrated Circuit Design	
	Credit Hours: 3	Contact Hours (L, H): (2, 2)
	Prerequisites: EE312	

Analysis and simulation of elementary transistor stages, current mirrors, supply- and temperature-independent bias, and reference circuits. Overview of integrated circuit technologies, circuit components, component variations and practical design paradigms. Differential circuits, frequency response, feedback and analog-to-digital conversion techniques will also be covered. Performance evaluation using computer-aided design tools.

EE365	Power Electronics	
	Credit Hours: 3	Contact Hours (L, H): (2, 2)
	Prerequisites: EE112, EE312	

This course covers topics which include characteristics of power semiconductor devices; Half-wave and fullwave rectifiers; AC voltage controllers, Static VAR control; DC-DC converters, Buck, Boost, and Buck-Boost design considerations; DC Power supplies, Transformer models, Flyback, forward, and push-pull converter, Half-bridge inverters, Full-bridge inverters, Multilevel inverters. Tutorials supplement the lectures by providing exercises and computer aided simulations to enhance the course understanding. The course is accompanied with laboratory sessions where student gain hands on experience in topics such as: Buck, Boost converters, Half and Full-bridge rectifiers, AC voltage controllers, Half and Full-bridge inverters.

EE401	Self study	
	Credit Hours: 2	Contact Hours (L, H): (0, 4)
	Prerequisites: None	

In-depth study of material not ordinarily covered in a course, under the supervision of a faculty member. Emphasis is on comprehension and presentation and survey of a clearly identified sub-area of electronics and/or communication engineering.

EE402	Minor Project	
	Credit Hours: 3	Contact Hours (L, H): (0, 6)
	Prerequisites: None	

Provides students with an opportunity for developing and implementing a system.

EE403	Minor Project	
	Credit Hours: 2	Contact Hours (L, H): (0, 4)
	Prerequisites: None	

Provides students with an opportunity for developing and implementing a system.

EE404	Minor Project	
	Credit Hours: 1	Contact Hours (L, H): (0, 2)
	Prerequisites: None	

Provides students with an opportunity for developing and implementing a system.

EE410	teDigital Signal Processingxt	
	Credit Hours: 3	Contact Hours (L, H): (2, 2)
	Prerequisites: EE201	

Sampling and quantization. Discrete time signals and systems. Difference equation models. Z-transform and applications to linear time invariant systems. Discrete-time Fourier transform. Discrete Fourier transform. Fast Fourier transform. Basic digital filter structures. IIR digital filter design. FIR digital filter design. Decimation and interpolation. Applications of digital signal processing in communication systems.

EE420	VLSI Design, Tools and Technology	
	Credit Hours: 3	Contact Hours (L, H): (2, 2)
	Prerequisites: EE212	

The MOS transistor. CMOS and Pseudo NMOS inverters. Pass transistors. CMOS circuits. CMOS technology: layout design, layout generation/partitioning, placements, and routing. Project involving design on-chip and simulation.

EE441	Wireless Communications	
	Credit Hours: 3	Contact Hours (L, H): (3, 0)
	Prerequisites: EE352	

The following topics will be covered; (1) History of wireless communications and specifications of different generations of cellular wireless communications. (2) Wireless communications channel, pathloss models, Friis equation, shadowing effect, different kind of fading, etc. (3) Spectrum Allocation: Licensed and unlicensed bands. (4) Cellular communications, Frequency reuse, cluster and capacity, hexagonal cells, co-channel interference, SIR, Sectoring. (5) Traffic handling capacity in wireless communications, Erlang B formula, notion of Trunking, tradeoff between Trunking efficiency and sectoring. (6) Duplexing and Multiple access schemes, TDD vs, FDD, FDMA and TDMA, CDMA and SDMA. (7) Spread Spectrum Communications, DS/SS, Pseudorandom sequence and m-sequences, cyclic codes, Sync. CDMA and orthogonality, Walsh sequences and Hadamard matrix, IS95-.

EE442	Optical Communications	
	Credit Hours: 3	Contact Hours (L, H): (3, 0)
	Prerequisites: EE341	

Review of optical sources and optical configurations. Noise sources. Optical live coding. Performance evaluation of optical receivers. Diversity receivers. Fiber link design. Introduction to optical space communication.

EE443	Wireless Networks and Protocols	
	Credit Hours: 3	Contact Hours (L, H): (3, 0)
	Prerequisites: EE341	

Study of theoretical and practical aspects of next-generation cellular and Wi-Fi communication systems including design principles, system and service requirements, implementation limitations and deployment scenarios using examples from real-life systems. Topics include radio access and core network protocols; centralized and distributed network architectures; power, mobility, and interference management; RF spectrum utilization; multiple-access schemes and multi-antenna transmission techniques; modern RF transceiver architectures and baseband signal processing and future trends in wireless communication.

EE444	Information Theory	
	Credit Hours: 3	Contact Hours (L, H): (3, 0)
	Prerequisites: EE341	

Measure of information, source coding and communication channel models. Channel capacity and coding: block codes, cyclic codes, BCH codes, Reed Solomon codes and convolution codes. Trellis coded modulation. Introduction to cryptography.

EE451	Image Processing	
	Credit Hours: 3	Contact Hours (L, H): (2, 2)
	Prerequisites: EE410	

-2D signals and systems. Image digitization. Image transforms. Image and video compression. Image enhancement. Color processing. Image restoration. Image segmentation. Multi-spectral analysis. Morphological processing.

EE452	Mechatronics	
	Credit Hours: 3	Contact Hours (L, H): (2, 2)
	Prerequisites: EE312, EE351	

Mechatronics: definitions and terminology. Mechanical elements with integrated electronics. Mechanics with integrated electronics. Precision mechanics.

EE453	Robotic and Automation	
	Credit Hours: 3	Contact Hours (L, H): (2, 2)
	Prerequisites: EE351	

Basic components of robotics systems. Kinematics of manipulators. Selection of coordinate frames. Solution of kinematics and manipulator dynamics path planning: velocity, force, computed torque controls. Linear and nonlinear controller designs for robots. Application in manufacturing and programmable automation.

EE467	RF Microelectronics	
	Credit Hours: 3	Contact Hours (L, H): (2, 2)
	Prerequisites: EE363	

Design of Radio Frequency (RF) integrated circuits for communications systems, primarily in CMOS. Topics comprise the design of oscillators, including PLLs and PLL-based frequency synthesizers, matching networks, low-noise amplifiers, mixers, modulators, and demodulators. Also, the design of high-efficiency (e.g., class E, F) RF power amplifiers, coupling networks.

EE470	Advanced Photovoltaics	
	Credit Hours: 3	Contact Hours (L, H): (2, 2)
	Prerequisites: EE365	

This course covers a wide range of materials. The first part of the course deals with the characteristics of sunlight, solar cells, and modules, with an emphasis on silicon solar cells. Students first become familiar with the properties of sunlight necessary to enable them to undertake PV systems design. The course then examines cell and module interconnection and their effect on performance. The second section of the course covers PV systems, predominantly on developing a technical understanding of PV system components and design. The focus is on stand-alone and grid-connected PV systems

EE481	Medical Electronics	
	Credit Hours: 3	Contact Hours (L, H): (3, 0)
	Prerequisites: EE312, EE201	

This course covers topics which include introduction to Lab-on-a-chip, Biosensors and Transducers, Electrochemical Probes, Bioinstrumentation and Buffer Amplifier, Bio signal Processing Linear Circuits, Bio interfacing and Control Circuits, Monitoring and signal display circuits.

EE491	Major Project, Part I	
	Credit Hours: 4	Contact Hours (L, H): (0, 8)
	Concurrent Prerequisites: GE480 and (EE365 or EE352) Minimum number of completed credit hours: 90	

This project-based course represents the first half of a project that requires students to demonstrate technical and practical skills and experience in some aspects of computer science/engineering, communications, and electronics engineering, as well as personal attributes at levels which commensurate with professional and ethical engineering practices. Students (Normally a group of 3 supervised by a KCST Supervisor) are normally assigned a project. Then they are expected to state its objectives, complete a literature survey, perform a feasibility study, set project specifications, and select a design method. Preliminary modeling and analysis are also expected from students to ensure that the necessary material needed for the completion of the project can be acquired in time for the completion of the project in the spring term. Teams will submit a professional report and do the relevant oral presentation. This course sequence prepares students for entry to the workplace. Deliverables include a Project Plan Document, a Design Document, and an oral presentation of project achievements.

EE492	Major Project, Part II	
	Credit Hours: 6	Contact Hours (L, H): (0, 12)
	Prerequisites: EE491	

This project-based course is a continuation of EE491 (FYPI). Students are asked to deliver a product that has passed through the design, analysis, testing and evaluation stages. Verification and validation must be applied at all stages, and implementation and testing must reflect that the design specifications including identified engineering ethics have been satisfied. Students are also expected to submit a final professional report that includes a description of the design process, implementation and testing, verification and validation and a critical appraisal of the project. A Final Report, an Oral Presentation of project achievements, and a Poster are also part of the project deliverables.

General Engineering Courses

GE480	Ethical, Legal and Professional Issues for Engineers	
	Credit Hours: 2	Contact Hours (L, H): (2, 0)
	Minimum number of completed credit hours: 60	

This course covers topics which include responsibility in engineering; framing the moral problem; organizing principles of ethical theories; computers; individual morality, and social policy; honesty, integrity, and reliability, safety, risk, and liability in engineering; engineers as employees; engineers and the environment; international engineering professionalism; and future challenges.

Languages Courses

LA101	English I	
	Credit Hours: 3	Contact Hours (L, H): (3, 0)
	Prerequisites: None	

LA101 is an Intermediate (Oxford-verified to be CEFR level B2) 3-credit course. This course combines thought-provoking videos and complex texts that give way to different tailored exercises that enhance student's critical thinking, writing and reading skills. Students will be able to formulate their own opinions and express themselves effectively with the use of appropriate and related terminology.xt

LA102	Arabic I	
	Credit Hours: 3	Contact Hours (L, H): (3, 0)
	Prerequisites: None	

An intensive introduction course to the Arabic language, focusing on reading and oral comprehension and speaking skills.

LA103	English II	
	Credit Hours: 3	Contact Hours (L, H): (3, 0)
	Prerequisites: LA101	

LA 103 is an upper intermediate (Oxford-verified to be CEFR level B2 – C1) credit course. This course is an integrated skills course that develops language and real academic skills essential for university level. The course includes lectures and seminars, language informed by Cambridge Academic Corpus and the Academic Word Lists, which guarantee that students are learning English that is up-to-date and relevant to them. LA 103 utilizes group work activities and individual assignments with the use of lecture skills. This course also emphasizes on training the students to use critical thinking techniques and developed independent learning skills. Within the intensive units, students will also review basic grammar and writing skills.

LA201	English for Engineers	
	Credit Hours: 3	Contact Hours (L, H): (3, 0)
	Prerequisites: LA101	

LA 201 is an advanced (Oxford-verified to be CEFR level C1 – C2) credit course. This course is an elective which provides students with the communication and written skills necessary to pursue career goals. This course is designed to introduce students to key technical issues utilized in today's professional fields. It aims to equip them with the language, research, reading and writing skills and strategies that are necessary to identify relevant information in the genres common in their field. Also, it aims to help students compose various technical documents tailored for a professional setting, including warning/ instruction emails, conceptual writing, describing graphs, and short report. Exposes students to a wide variety of technical language skills to use in technical writing tasks. Assists students to engage in and demonstrate critical thinking skills such as analyzing, evaluating, constructing, and supporting an argument to produce technical documents. Help students adopt an integrated approach that develops written and presentation skills within engineering contexts.

Mathematics Courses

MA101	Mathematics I	
	Credit Hours: 3	Contact Hours (L, H): (3, 0)
	Prerequisites: None	

This course is an introductory calculus course. It is a part of a three-course sequence and precedes Calculus II and III. Topics include functions of a single variable, limits and continuity, derivatives, applications of differentiation, integrals, and areas between curves.

MA102	Mathematics II	
	Credit Hours: 3	Contact Hours (L, H): (3, 0)
	Prerequisites: MA101	

This course covers topics which include different techniques of integrations, improper integrals, infinite sequences and series, power series representations of functions, parametric equations, and polar coordinates.

MA111	Discrete Mathematics	
	Credit Hours: 3	Contact Hours (L, H): (3, 0)
	Prerequisites: None	

This course introduces basic concepts of discrete mathematics used in computer science and engineering disciplines that involve formal reasoning. The topics include logic, sequences, mathematical induction, and recursion, set theory, functions and relations, and counting.

MA203	Linear Algebra	
	Credit Hours: 3	Contact Hours (L, H): (3, 0)
	Prerequisites: None	

This course introduces the basic concepts and techniques of elementary linear algebra. Topics include systems of linear equations, Gaussian elimination, matrix algebra, inverse and determinant of square matrices, vectors in n-space, vector spaces and subspaces, span and linear independence, basis and dimension, eigenvalues and eigenvectors, and diagonalization of a matrix.

MA204	Engineering Mathematics	
	Credit Hours: 3	Contact Hours (L, H): (3, 0)
	Prerequisites: MA102	

The course focuses on the multivariable calculus and mathematical methods needed to understand real-world problems involving quantities changing over time in two and three dimensions. This course covers differential, integral, and vector calculus for functions of more than one variable, important mathematical foundations for advanced studies in physical sciences, engineering, computer graphics, social sciences, and tools to describe and understand phenomena in the engineering world as well as improving analytic and problem-solving skills valuable in many disciplines.

MA213	Computational Linear Algebra	
	Credit Hours: 3	Contact Hours (L, H): (2, 2)
	Prerequisites: MA203	

This course provides an overview of numerical methods for solving large-scale linear algebra problems arising in science and engineering. Topics include Gauss elimination, Doolittle's and Choleski's methods, iterative methods of Jacobi, Gauss-Seidel, SOR, conjugate gradient, and GMRES, linear least squares problem, and matrix eigenvalue problem.

MA301	Probability and Statistics	
	Credit Hours: 3	Contact Hours (L, H): (3, 0)
	Prerequisites: MA102	

Probability theory: Algebra of sets, sample space, independence, conditional probability, Bayes' theorem. Random variables: discrete and continuous probability distributions, moments. Joint probability: conditional distributions, moments, product moments. Statistical inference. Estimation: Point and interval estimation, hypothesis testing. Regression analysis: Linear and non-linear.ext

MA302	Ordinary Differential Equations	
	Credit Hours: 3	Contact Hours (L, H): (3, 0)
	Prerequisites: MA102	

This course covers topics which include first order differential equations, linear equations of second and higher order, Laplace transforms with applications, series solutions differential equations with variable coefficients

MA315	Operations Research	
	Credit Hours: 3	Contact Hours (L, H): (3, 0)
	Prerequisites: MA301	

Basic methodology and techniques of operations research. Emphasis on application and problem-solving models; linear programming, sensitivity analysis, nonlinear/classical optimization, queuing theory; Markov processes; dynamic programming.

MA316

Complex Variables	
Credit Hours: 3	Contact Hours (L, H): (3, 0)
Prerequisites: MA203, MA204	

The course starts with a quick reminder of The Algebra of Complex Numbers. Then, it addresses topics like analyticity, Cauchy-Riemann Equations, harmonic Functions, exponential, trigonometric and inverse trigonometric, hyperbolic, and logarithmic Function. It also focuses on Contours Integrals, independence of path, and Cauchy's Integral Theorem and Formula. Topics like Taylor series, power series are revisited but with complex variables, and extended to Laurent Series in various domains of the complex plane. The residue theorem is addressed and applied to trigonometric integrals and improper integrals of certain functions. This course is very important for students getting involved in research projects. Its applications cover a wide range of engineering applications.

MA317

Number Theory	
Credit Hours: 3	Contact Hours (L, H): (3, 0)
Prerequisites: MA111	

Divisibility of integers. Linear and nonlinear Diophantine equations. Prime numbers. The fundamental theorem of arithmetic. The prime number theorem. Famous unsettled Conjectures about prime numbers. Irrationality of e, , and 2 . Transcendental numbers. Congruences. Fermat's and Wilson's theorems. Euler's function and theorem. Fermat's last theorem and Pythagorean triples. Law of quadratic reciprocity.

MA318

Numerical Methods for Engineers	
Credit Hours: 3	Contact Hours (L, H): (2, 2)
Prerequisites: MA203, MA204	

This course creates, analysis, and implements algorithms for finding numerical solutions for various problems arising throughout the natural sciences and engineering, especially for those in which analytical solutions do not exist. Topics include numerical solution of nonlinear equations, numerical solution of systems of linear equations, polynomial interpolation, Least-Square approximation, and numerical differentiation and integration.

MA340

Abstract Algebra	
Credit Hours: 3	Contact Hours (L, H): (3, 0)
Prerequisites: MA203	

Semi-groups, monoids, groups, rings, fields, ideals, modules. Fundamental theorems (including isomorphism, Sylow's theorem, etc.) for these algebraic structures. Morphisms between algebraic structures. Galois theory. Introduction to category theory: categories, functors, natural transformations, adjoints

MA350

Topology and Functional Analysis	
Credit Hours: 3	Contact Hours (L, H): (3, 0)
Prerequisites: MA203	

Metric spaces, open, closed and boundary sets. Closure, convergence, and completeness. Bolzano-Weierstrass theorem. Topological spaces, continuous functions and homomorphisms. Hausdorff-Tychonoff normal spaces. Compactness, connectedness, separability. Normed linear spaces, Banach spaces. Hahn-Banach theorem. Open maps and closed graphs. Hilbert spaces. Orthogonal complements, direct sums, orthonormal sets. Reisz's representation theorem, self-adjoint, unitary and normal linear operators.

Physics Courses

PH101	Physics I	
	Credit Hours: 4	Contact Hours (L, H): (3, 2)
	Prerequisites: None	

This is an introductory course on principles and methods of physics in classical Mechanics and waves. This course starts from the concepts of scalar and vector physical quantities, mathematical operations of vectors, and both scalar and vector products. Then, the course covers the kinematics in one-dimension, constant acceleration motion, free falling, two/three-dimension kinematics, and projectile motion. Then, the course focuses on Newton's laws of motion and its engineering applications, relation between work and force, concepts of kinetic and gravitational potential energies, conservation of energy principle, and elastic potential energy. In addition, the course covers rotational motion of rigid bodies, gravitational forces, gravitational potential energy, satellite motion, and Kepler's law to describe the motion of planets. Finally, the course presents an introduction about wave mechanics and simple harmonic motion.

PH102	Physics II	
	Credit Hours: 4	Contact Hours (L, H): (3, 2)
	Prerequisites: PH101	

This is an introductory course on principles and methods of physics for students who have good preparation in Mechanics. This course covers electricity and magnetism using calculus-based approach. Electromagnetism: Coulomb's law, electric fields: electric field of a point charge, electric field of a collection of charges, electric field due to continuous charge distributions, Gauss's law: electric field of a charged conducting sphere, electric field of a sheet of charge, electric potential: electric potential energy of a system of charges, potential energy associated with a point charge, potential at a point due to collection of charges. potential due to ring of charge, potential due to a line of charge, potential due to a sphere, equipotential surfaces, capacitance: parallel plate capacitor, capacitors in series and parallel, dielectrics, current, current density, drift velocity, resistivity, Ohm's law, resistance, electromotive force, internal resistance of a battery, free electron theory of metallic conduction, resistors in series and parallel, Kirchoff's rules, charging and discharging of a capacitor in RC- circuits, magnetic fields, Lorentz force, magnetic force on current carrying wire, DC motor, Hall effect, magnetic field due to currents, magnetic field due to circular current loop, Ampere's law, Faraday's law, Lenz's law, generator, motional electromotive force, induced electric fields.

PH103	Solid State Physics	
	Credit Hours: 3	Contact Hours (L, H): (3, 0)
	Prerequisites: PH102	

This course develops concepts in quantum mechanics such that the behavior of the physical universe can be understood from a fundamental point of view. This course will cover the following topics. Schrodinger equation, operators, eigenfunctions, observables, infinite well in one and three dimensions, tunnelling and quantum dots, harmonic oscillator potential, energy bands in solids, allowed and forbidden energy bands, electrical conduction in solids, fermions and bosons, statistical mechanics, probability distribution functions, Fermi-Dirac probability function, Fermi Energy and Fermi Level, Maxwell-Boltzmann approximation, and P-N Junctions.

PH201	Modern Physics for Engineers	
	Credit Hours: 3	Contact Hours (L, H): (3, 0)
	Prerequisites: PH102, MA102	

This course offers an introduction to modern physics with emphasis on quantum physics. Thus, it lays the foundation for later more advanced courses that deal with applications of quantum mechanics to condensed matter physics, solid state devices, materials science & engineering. The course begins with a review of the experimental basis of quantum mechanics & the structure of the atom. This is followed by a discussion of the wave properties of matter & quantization. Next probability, wave functions & the Copenhagen interpretation of quantum mechanics are presented. The final part of the course focuses on detailed applications of the Schrodinger equation to simple systems & the hydrogen atom

PH301	Solid State Physics and Devices	
	Credit Hours: 3	Contact Hours (L, H): (3, 0)
	Prerequisites: PH201, PH103	

This course will build on the basics of PH201 to introduce the physics of semiconductor materials. The main target is to integrate different topics together including both semiconductor material physics and semiconductor device physics. Therefore, the course will help students to understand the operation of present day's semiconductor devices and future developments in the field. The present course is designed to help students develop a much better intuition about physical phenomena, especially understanding concepts and problem-solving strategies. Moreover, the latest updates in semiconductor fabrication techniques (FAB) will be presented in this course, which gives an experimental/industrial flavor of the course for the undergraduate students. Progress towards these outcomes will be assessed through in-class exams, homework assignments, quizzes, and a possible mini-review paper as a project and discussion exercises. In some details, our course is mainly divided into three parts. In the first part, PH202 course initially presents a quick review about quantum mechanics concepts, such as energy quanta and wave-particle duality. Then, the course will review Schrodinger's Wave Equation and its applications in different potential environments. Moving to Kronig-Penny Model, the students will be able to understand the origin of bandgap and know the difference between semiconductors and other states of solids. Then, different probability density functions are presented ending with Fermi-Dirac one, which acts as a real start of semiconductor world within both intrinsic and extrinsic cases and its related devices. In the second part of the course, a connection between the concepts of semiconductors and the devices will be established. Different devices will be presented in detail such as pn diode, Bipolar Junction Transistor, Junction Field Effect Transistor, Metal-Semiconductor and Metal Oxide-Semiconductor Field Effect Transistor. Within each device, it will be investigated the full physical concepts of operation, characteristics, and its circuit modeling. Over the first two parts of the course, a lot of numerical examples will be presented to enrich the physical explanation of the concepts whether for the materials or the correlated devices. The third part of the course will briefly present an overview about semiconductor fabrication procedure in the cleanroom.

Social Sciences Courses

SS101	History of Arab Civilization	
	Credit Hours: 3	Contact Hours (L, H): (3, 0)
	Prerequisites: None	

In this course, students will be introduced to several subjects related to the evolution and development of Arab culture. Special attention will be devoted to the history of the Arab region and its civilizations, of the Arab people and their language, religion and culture. Upon completion of this course, students will be able to identify the key aspects of Arab-Islamic culture. They will also have a solid knowledge of the Islamic civilization and the contribution of Arab Scholars to global civilization and science.

SS104	History of Kuwait	
	Credit Hours: 3	Contact Hours (L, H): (3, 0)
	Prerequisites: None	

This course is a model of the expanded historical studies that present a new vision in its handling of the history of the State of Kuwait, as the book relied on the balance between the vision and the ancient classical text based extensively on the primary historical sources and literature in addition to those contemporary academic studies and research, which allows the student to form A clearer and broader vision of the reality of the event and how to monitor it from various angles and orientations, through multiple fields of knowledge such as history, politics, international relations and sociology.

This course shows the history of the emergence of the State of Kuwait and its establishment at the hands of Al- Utub, and the most important events that occurred in Kuwaiti society. So, the course based on in a logical manner in the narration of its political history of Kuwait, in terms of how to write Kuwaiti history by relying on reliable sources after it moved from the stage of transmitting oral narrations to writing and documenting them.

One of the important topics in this course is that it discusses the true history of the founding of the State of Kuwait with evidence and proofs despite the conflicting opinions about it, and the ambiguity that surrounds it when establishment. Also, it explains the main role for each ruler and his own imprint internally and externally warding off the various dangers from Kuwait by thanks to their wise policy. That means the course clarifies the external problems that the State of Kuwait suffered.

Finally, it is important to talk about the features of Kuwaiti society, and to clarify the most important ancient marine activities practiced by Kuwaitis, as well as their social, educational, health and judicial activities.

SS111	Introduction to Economics	
	Credit Hours: 3	Contact Hours (L, H): (3, 0)
	Prerequisites: None	

This course is aimed at introducing the principles of economics to students of science and engineering at an early stage of their undergraduate studies. The key objective is to give students a broad understanding of the forces shaping the competitive markets, both from a macro and a micro perspective. The course will introduce the key topics of microeconomics, including the market forces of supply and demand, costs of production, the markets for the factors of production and types of market structures, other than open competitive markets, namely monopolies and oligopolies. Basic macroeconomic concepts, such as national income, cost of living, unemployment, monetary system will also be introduced to the students, but always linked to their microeconomic counterparts.

SS212	Macroeconomics	
	Credit Hours: 3	Contact Hours (L, H): (3, 0)
	Prerequisites: SS111	

Classical and Keynesian theory on employment and economic activity. The macroeconomics of open economies. Aggregate demand and aggregate supply. Influence of monetary and fiscal policy on aggregate demand. Short-run trade-off between inflation and unemployment. An important component of this course will be Islamic finance.

SS221	Introduction to Psychology	
	Credit Hours: 3	Contact Hours (L, H): (3, 0)
	Prerequisites: None	

History of Psychology. Branches of psychology and its relevance in life. Biological Foundations. Sensory process. Perception: attention, form, depth, movement, constancy of perception. Learning and memory. Principles of learning, forgetting. Emotions, motives and motivation. Development processes. Intelligence. Thinking: processes, problem-solving, decision making. Personality, behavior, abnormal behavior and disorders.

SS231	Philosophy and Ethics	
	Credit Hours: 3	Contact Hours (L, H): (3, 0)
	Prerequisites: None	

Principle of philosophy. Morality, goodness and virtue. Moral imperatives and obligations. Ethical theories: utilitarianism, Kantianism.

SS332	The Philosophy of Science	
	Credit Hours: 3	Contact Hours (L, H): (3, 0)
	Prerequisites: None	

Perceptual knowledge. Empiricism. The scientific method. Observation, induction, deduction. Falsifiability of hypotheses. Theories of knowledge.

Foundation Courses

Information Technology Courses

IT001	Computer Skills	
	Credit Hours: 0	Contact Hours (L, H): (2,0)
	Prerequisites: placement test results	

The goal of this very hands-on course is to introduce the student to computer terminology, concepts and skills. It opens the various possibilities of applying information technology in everyday life and making effective use of office and personal applications, computing and communication devices and digital media in general.

IT002	Introduction to Information Technology	
	Credit Hours: 0	Contact Hours (L, H): (2,0)
	Prerequisites: IT002 or placement test results	

The goal of this very hands-on course is to introduce the student to the professional use of information technology and, in particular, Microsoft Office tools for making effective and professional reports and presentations. It aims at raising the standard of using IT in writing reports and essays as well as presenting their work using advanced presentation techniques that include multimedia.

Languages Courses

LA001	Pre-Foundation English	
	Credit Hours: 0	Contact Hours (L, H): (3,0)
	Prerequisites: placement test results	

LA 001 is a pre-elementary level (Oxford-verified CEFR Level A1) non-credit course. The purpose of this course is to build on existing basic knowledge of the English language in terms of vocabulary, reading, writing, and speaking skills. Students will revise their understanding of simple English sentences, structuring sentences, and building a vocabulary. LA 001 utilizes group work activities, individual assignments, and presentation skills.

LA002	Foundation English Level II	
	Credit Hours: 0	Contact Hours (L, H): (3,0)
	Prerequisites: LA001 or placement test results	

LA 002 is an elementary level (Oxford-verified to be CEFR level A2) non-credit course. This course is designed to provide English Language instruction to non-native students who need to improve their English language and academic skills in preparation for their undergraduate studies at KCST. This course addresses a complete range of language skills, including reading, writing, academic listening, oral communication, vocabulary, grammar, and critical thinking through a learner-centered approach. At this level, students will master the basics of English and acquire the necessary academic skills they need to progress to English Foundation LA 003 and then transition to their undergraduate degree programs.

LA003	Foundation English Level III	
	Credit Hours: 0	Contact Hours (L, H): (3,0)
	Prerequisites: LA002 or placement test results	

LA 003 is an intermediate level (Oxford-verified to be CEFR level B1) non-credit course. This course is designed to provide English Language instruction to non-native students who need to improve their English language and academic skills in preparation for their undergraduate studies at KCST. This course addresses a complete range of language skills, including reading, writing, listening, oral communication, vocabulary, grammar, and critical thinking through a learner-centered approach. At this level, students will build upon what they learned at the previous level, LA 002. They will develop intermediate English and improve their academic skills to transition to credit-bearing Academic English courses and their undergraduate degree programs.

Mathematics Courses

MA002	Foundation Mathematics Level II	
	Credit Hours: 0	Contact Hours (L, H): (3,0)
	Prerequisites: placement test results	

This course covers topics which include Real numbers, Exponents and radicals, Algebraic and Rational expressions, Equations, complex numbers, Inequalities, lines, functions and their graphs, Transformation of functions, combining functions, one-to-one functions and their inverses

MA003	Foundation Mathematics Level III	
	Credit Hours: 0	Contact Hours (L, H): (3,0)
	Prerequisites: MA002 or placement test results	

This course covers topics which include quadratic and polynomial functions, dividing polynomials, real zeros of polynomials, rational functions, polynomial and rational inequalities, logarithmic functions, laws of logarithms, exponential and logarithmic equations, trigonometric functions, trigonometry of right triangles, trigonometric functions of angles, trigonometric identities, addition and subtraction formulas, double angle formulas.

Physics Courses

PH002	Foundation Physics Level II	
	Credit Hours: 0	**Contact Hours (L, H): (3, 0)**
	Prerequisites: placement test results	

Physical quantities, dimensions and units of measurement. Algebra and solution of simultaneous equations, vector algebra. Kinematics in one dimension. Free fall. Basics of thermal physics.

PH003	Foundation Physics Level III	
	Credit Hours: 0	**Contact Hours (L, H): (3,0)**
	Prerequisites: PH002 or placement test results	

This course comprises the second part of the foundation physics course of KCST. Emphasis is on the fundamental laws of nature describing motion, explanation of motion using forces and Newton's laws, looking for what stays the same when everything else is changing (momentum and energy), and description and explanation of wave motion and static electricity. Considerable emphasis is on problem-solving and evidence-based decisions making.